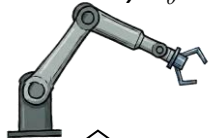


## Trajectory Rollout

### Instruction

Pick up the black bowl next to the ramekin and place it on the plate

VLA Policy  $\pi_\theta$



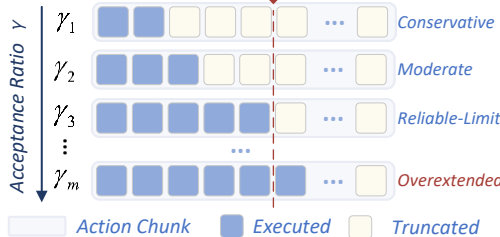
### Observation



## Stage1: Reliable Action Chunk Extension

### Dynamic Execution Horizon Exploration

Reliability Frontier



### Reliable Horizon Reward



Adv

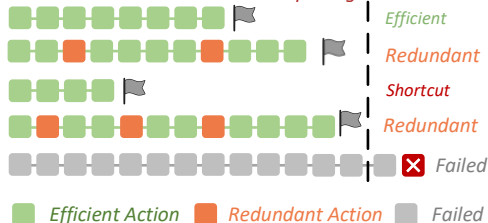
Update

Init  $\pi_\theta$  and  $\gamma$  form Stage1

## Stage2: Redundancy-Aware Step Reduction

### Variable-Length Step Trajectories

Step Budget



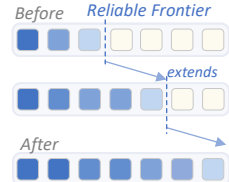
### Step Efficiency Reward



Adv

Update

## Progressive Horizon Expansion



Execution Window ↑

Policy Efficiency

Execution Trajectory ↓

### Redundant Step Elimination

